Reg. No. | | | | | | | | | | |

[14PH12/22]

FIRST SEMESTER B.TECH EXAMINATION, DECEMBER - 2015

ENGINEERING PHYSICS

Time:3 Hours **Maximum Marks:70**

Instructions : Answer any **FIVE** questions choosing at least TWO questions from each part.

List of constants:

i. **Velocity of light, C = 3×10^{8} m/s**

ii. **Planck's constant, h = 6.626×10^{-34} J.S**

iii. **Mass of electron, m_e = 9.1×10^{-31} Kg**

iv. **Charge of electron = 1.6×10^{-19} C**

v. **Mass of neutron, m_n = 1.67×10^{-27} Kg**

vi. **Permittivity of Vaccum ϵ_0=8.85×10^{-12} F/m**

vii. **Boltzmann constant, K = 1.38×10^{-23} J/K**

1. a. Explain phase velocity and group velocity. **02**

   b. Describe Davission Germer experiment and explain how it established the proof for wave nature of electrons. **08**

   c. Calculate the de Broglie wavelength associated with 400g cricket ball with a speed of 90 Km/hr. **04**

2. a. Explain the terms : i. Spontaneous emission **02**
   ii. Stimulated emission

   b. Explain the construction and working of a He-Ne laser. **08**

   c A laser beam with power per pulse is 1 mW lasts 10 ns, if the number of photons emitted per pulse is 3.9×10^{7}, calculate the wavelength of laser. **04**

3. a. Explain the principle and working of optical fibers. **03**

   b. What is attenuation? Explain the various mechanism through which attenuation takes place. **07**

   c. The numerical aperture of an optical fibre is 0.2 when surrounded by air. Determine the R.I of its core, given the R.I of the cladding is 1.59. Also find the acceptance angle when the fibre is in water of R.I 1.33. **04**

4. a. Distinguish between hard and soft magnetic materials. **06**

   b. Explain magnetic hysteresis on the basis of domain theory. **05**

   c. In a magnetic material, the field strength is found to be 10^{6} Am^{-1}. If the magnetic susceptibility of the material is 0.5×10^{-5}, calculate the intensity of magnetization and the flux density of the material. **03**

SCHOOL OF ENGINEERING & TECHNOLOGY LIBRARY DATE JAIN UNIVERSITY BANGALORE

PART-B

5. a. Define packing factor. Calculate the packing factor for SC, BCC and FCC structures. **06**

   b. Describe the construction and working of a Bragg's X-ray spectrometer. **05**

   c. Draw the following planes in a cubic unit cells: **03**

      i. (200) ii. $(\bar{2}00)$ iii. $(1\bar{3}2)$

6. a. Explain screw dislocation and edge dislocation with a neat diagram. **05**

   b. Explain the crystal structure of NaCl with a neat diagram. **06**

   c. Write a note on grain boundaries. **03**

7. a. Explain electronic polarization. **03**

   b. Derive an expression for internal field in liquid or solid dielectric for one dimensional. **08**

   c. A parallel plate capacitor consist of 2 plates each of area $5\times10^{-4}m^2$. They are separated by a distance $1.5\times10^{3}m$ and filled with a dielectric of relative permittivity 6. Calculate the charge on the capacitor if it is connected to a 100 V dc supply. **03**

8. a. What are nano materials? Explain three types of nano materials. **04**

   b. Explain with a neat diagram of synthesis of nano materials by using Ball milling method. **07**

   c. Explain any three physical properties of nano materials. **03**